Attention-Aware Sobel Graph Convolutional Network for Remote Sensing Image Change Detection

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***Abstract*— In the study of remote sensing images, the problem of change detection (CD) is crucial. Convolutional neural networks (CNNs) are well-liked feature extraction structures that are frequently used in CD. On the other hand, graph convolutional networks (GCNs) are effective in building contextual structure information. Compared with CNN, GCN can make full use of the graph structure information to capture the changing features between different areas in the graph by learning the connections and interactions between nodes. In contrast, traditional pixel-based CNNs may have difficulty modeling semantic relationships and temporal variations among features and are susceptible to noise interference. So in this article, we extract optimization information using a GCN structure. Due to the particularity of remote sensing images, edge information is often ignored, which is useful in the field of CD. In this article, we propose an attention-aware Sobel GCN (ASGCN) for remote sensing image CD. First, we use a Siamese CNN to extract primary multilevel features. Then, a dual-branch attention module (DAM) including coordinate attention and multiscale local attention module (MLAM) is proposed to focus on informative pixels, we use Sobel operator to construct graph, and the graph convolutional module can expand receptive field and extract edge information. Attention fusion module (AFM) is adopted at decoder to perform effective feature fusion. Extensive comparative experiments on three CD datasets, LEVIR-CD, WHU-CD, and DSIFN-CD, verify the effectiveness of the proposed ASGCN.**

***Index Terms*— Attention-aware, change detection (CD), graph convolutional network (GCN), remote sensing image, Sobel operator.**

I. Introduction

CHANGE detection (CD) is a technology in remote sensing images which sets the pixel to 0 or 1 in multitemporal images and the same area to determine whether the area has changed or not [1], and the technology can be applied to practical problems such as ecological research [2], city planning [3], and damage evaluation [4]. As satellite imaging technology has developed quickly, numerous multitemporal remote sensing images have been collected by a variety of platforms, including satellites and aircraft [5]. It increases the need for intelligent CD technology and gives the CD field more precise data. For CD, two or more photographs of the same geographic area are taken at various periods. Owing to the unique qualities of remote sensing images such as specificity of view, scale, and small targets, there are challenges in performing CD in the field of remote sensing: lots of interfering information, complexity of objects in the scene, light angle, shadows, reflections from ground objects, soil washout, or other factors may affect the precision of CD.

Traditional methods in CD focused on multitemporal very high-resolution (VHR) can be classified into the following four groups: 1) image algebra methods; 2) classification-based methods; 3) transformation-based methods; and 4) other advanced methods.

Algebra-based methods [6] represent that change maps are obtained by performing algebraic operations. However, this method is sensitive to changes in illumination, which usually considers only the differences between pixel values, are more sensitive to changes in illumination. When there is a small change in brightness or color between two images, it may produce false or missed results.

Classification-based methods [7] determine the change in areas by distinguishing the differences between classified images, or directly detect changes in multitemporal remote sensing images using a trained classifier. While this approach has some advantages, it also has some disadvantages; it requires large amounts of labeled data, which is time-consuming and human-resource-intensive. Acquiring and labeling large-scale labeled datasets can be a challenge.

Another traditional CD method is the transform-based approach, which accomplishes CD by transforming the image into a new feature space to extract effective features and reduce data redundancy, in which changed features are highlighted and unchanged features are suppressed. However, this method is difficult to determine the threshold value, and it is a challenge to choose the appropriate threshold value to determine the change or not.

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Other advanced typical CD methods have been put forth. For instance, some researchers have used a physical model to convert the challenge of CD into a test of statistical hypotheses [13].

Remote sensing image CD methods which based on deep learning are gradually emerging as deep learning technology develops. These methods can automatically learn and extract feature information of change targets without manual parameter setting and have higher accuracy and robustness. Remote sensing images have characteristics such as large scale and high resolution, and the objects in the images generally have a certain scale, in which case the use of traditional image processing algorithms can easily lead to excessive smoothing or distortion. As AI techniques continue to evolve in the field of remote sensing imagery, many works are widely used. Hong et al. [14] propose a high-resolution domain adaptive network to improve the generalization of AI models in multicity environments and use a parallel high- and low-resolution fusion approach to well-preserve the spatial topology of the studied urban scenes. Hong et al. [15] designed SpectralGPT, a groundbreaking RS base model carefully tailored for spectral data. It features a 3-D masking strategy, an encoder for learning visual representations from spatial–spectral hybrid tokens and a multitarget reconstruction decoder for preserving spectral sequence features. Convolutional neural networks (CNNs) can effectively process spatial information and extract features at different scales when processing images. Processing a lot of data is necessary for CD in the field of remote sensing image, which is very computationally intensive if we use conventional machine learning techniques that demand human feature extraction. CNNs have the capacity to process lots of data using techniques such as image blocks and batch training, which dramatically improves computational efficiency.

Graph CNNs [45] have also demonstrated advancement in the realm of CD, alongside CNNs. Remote sensing images are mostly irregular image data, in which the shapes, sizes, locations, etc. of features and scenes are varied. Traditional CNNs can generally only handle regular grid data, while graph convolutional networks (GCNs) can easily handle arbitrarily complex image structures. Traditional CNNs usually consider only local information, which makes it challenged to integrate global information. CD needs to take into account the correlation between regions in the whole image. For GCNs, the core operation is on the graph nodes, and thus, the global relationship between nodes can be used to better extract the feature information.

The current method of CD in the field of remote sensing images is still not fine enough in dealing with irregular data, does not have better access to feature information in the global relationship, and has poor detection performance in the case of large background noise, so we designed an improved GCN to build a GCN using the Sobel operator to construct the adjacency matrix. The Sobel operator is a commonly used image edge detection operator that can be used to construct a sparse adjacency matrix to implement graph convolution operations. We use the Sobel operator to construct the adjacency matrix m which can construct the graph convolution adjacency matrix which also ensures the robustness and effectiveness of the model to a large extent.

The following is a list of this article's main contributions.

1) Aiming at the problems of the current deep learning models in the field of remote sensing image CD such as insufficient extraction of global context information and ignoring feature edge information, we propose a CD model based on attention-aware Sobel GCN (ASGCN). The model uses the Sobel edge operator to construct a GCN to extract global information and focus on edge features.
2) The attention mechanism is introduced and a two-branch attention module is proposed, in which the multiscale local attention module (MLAM) carries out feature refinement in multiple perspectives and focuses on the local information of features while extracting multiscale feature information.
3) At the same time, we propose an attention fusion module (AFM), which more efficiently fuses the encoder dual-branch feature information and decoder feature information.

We conduct a large number of comparative experiments and ablation experiments on three remote sensing image CD datasets (LEVIR-CD, WHU-CD, and DSIFN-CD) to verify the effectiveness of the model.

II. Related Works

In this section, we provide a brief review of CNN-based CD methods and GCNs, as they are related to this article.

A. CNN-Based CD

In recent years, deep learning (DL) has developed into an influential technology for a variety of computer vision tasks, and it is able to extract depth features to more accurately represent image information. In addition, RS image processing makes extensive use of it and related research has been conducted in several fields such as remote sensing segmentation [20], landslide detection [21], and image classification [22].

Automatic CD technology based on DL can reduce a lot of manual identification time and get good accuracy. Unlike traditional methods, CNN-based CD methods can extract multiscale features more effectively and refine features from different layers; it is a classical deep learning structure that makes it possible for the network to extract spatial–spectral features in an end-to-end way. Similarly, in the field of CD, CNNs have shown effective performance. As an example, it was first suggested to use a Siamese network to compare two image pairs. Multiple inputs can be used concurrently for comparing images using it, which is applicable for CD.

Recently, many researchers have proposed models for CD using different strategies. Zhan et al. [23] used Siamese networks in CD for the first time. Guo [24] proposed a new fully convolutional tandem metric network capable of metric change by customizing an implicit metric and proposed a threshold contrast loss method. To further improve the

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accuracy by optimizing loss function, STANet [25] and DASNet [26] further optimized weighted contrastive loss function and introduced batch-balanced contrastive loss function and weighted double-margin contrastive loss function, respectively. Peng et al. [27] first combined the multitemporal images as the input of the improved UNet++ to obtain multilevel feature maps, which are used to determine the change areas, and Peng et al. [28] used the attention mechanism in UNet++ to enhance the performance of the network. Similarly, Liu et al. [29] and Chen et al. [26] proposed a dual-attention module to optimize features in the direction of both channel and spatial. To diminish the convolution operation's inherent locality, Peng et al. [30] used the dilated convolution to increase the receptive field.

Remote sensing image CD needs to take into account the temporal changes in features and contextual information of their surroundings, when using pure convolution for remote sensing image CD; they are computationally inefficient and require high computational complexity.

B. Graph Convolutional Network

Compared with CNNs, GCNs have the ability to model irregular data structures and they are widely used in various vision tasks, such as image classification, object detection, and semantic segmentation. CNNs are good at processing regular graph data and represent images as a regular grid in the Euclidean space; it can take advantage of the shift in-variance and local connectivity of image data, so it is more meaningful when extracting local features. Past studies have proven its effectiveness in image processing.

However, GCNs are more universal for graph data, it can handle non-Euclidean data, the structural features of the image can be fully maintained, and the class boundaries are maintained more flexibly. New generalizations and definitions of convolutional operations have been rapidly developed in the past few years and can be used as a basis for GCNs, GCNs integrate and transform the feature information of the graph's nodes, and subsequently successfully extract the image's features of irregular regions using the constructed graph structure, which can be used to extract global image features and effectively model the complex spatial relationships of VHR. A growing number of experiments demonstrate the usefulness of graph convolution for specific functions in different domains.

GCN can easily handle arbitrarily complex image structures. In the field of remote sensing images, adaptive graph convolution network (AGCN) [31] uses an attention mechanism layer and GCNs, which generalize the convolution to the graph domain to deal with graph-structured data, and has been successfully applied to tasks such as node classification. Qin et al. [33] proposed a semi-supervised GCN for HSI classification, which the features of each pixel can be propagated along the spatial dimension based on their spectral similarity and spatial distance. The structural complexity of graphs imposes an expensive computational cost on large graphs, so Wan et al. [34] used super-pixels but not pixels as nodes in the graph structure, enabling GCNs to model the large-scale spatial structure of HSIs with an acceptable computation. Furthermore, Liu et al. [32] introduced a CNN-enhanced GCN that integrates CNNs and GCNs, with CNNs generating pixel-level features and GCNs generating super-pixel-level features. There is also a local convolution for fast approximation, and a simple hierarchical propagation rule is designed for semi-supervised classification, enabling the GCN structure to encode both node and graph features.

However, in previous methods, edge information is ignored. Changes in remote sensing images often occur at the edges of features or scenes, such as the edges of buildings, roads, or vegetation. Edge information can help the model to locate and identify these change areas more accurately. Edge information helps provide richer spatial context, enabling models to better understand the relationships between features. This is important for recognizing local variations and details, especially in high-resolution remote sensing images.

In the graph convolution algorithm, the adjacency matrix is used to calculate the feature information of each node with its neighbor nodes and generate a new node feature vector. By continuously iterating this process, the feature information of the graph can be gradually extracted to achieve tasks such as classification and clustering. For graph CNNs, information on the topology between nodes can be easily captured through the adjacency matrix.

Based on the help of edge information and GCN for CD, this study proposes a graph convolution constructed using Sobel edge operator to focus on edge information and improve CD accuracy.

III. Method

The overall structural framework of the proposed ASGCN-CD is shown in Fig. 1. We send the input multitemporal images into the two branches of the encoder. First, the input image is extracted by several convolutional layers, which is the feature extraction module. Then, it is divided into two branches into two attention modules, which are the coordinate attention module and MLAM. Then two branches are input to the ASG module; the graph is constructed by the Sobel operator in different directions, the edge features are extracted, and the features are refined. Use AFM to fuse the feature maps of different scales in both the encoder and decoder, then go through the decoder, and finally output the change map. In the following, the proposed ASGCN-CD is introduced in detail by analyzing the feature extraction (see Section III-A), introducing the dual-branch attention module (DAM) (see Section III-B), and explaining the GCN network with the Sobel operator (see Section III-C) and the encoder with the fusion module (see Section III-D).

A. Feature Extraction Module

We use 3×3 convolution in the feature extraction module to moderately reduce the number of model parameters to obtain more information of local features. It also increases the depth of the network to enhance the nonlinear feature learning ability and better adapt to data distribution.

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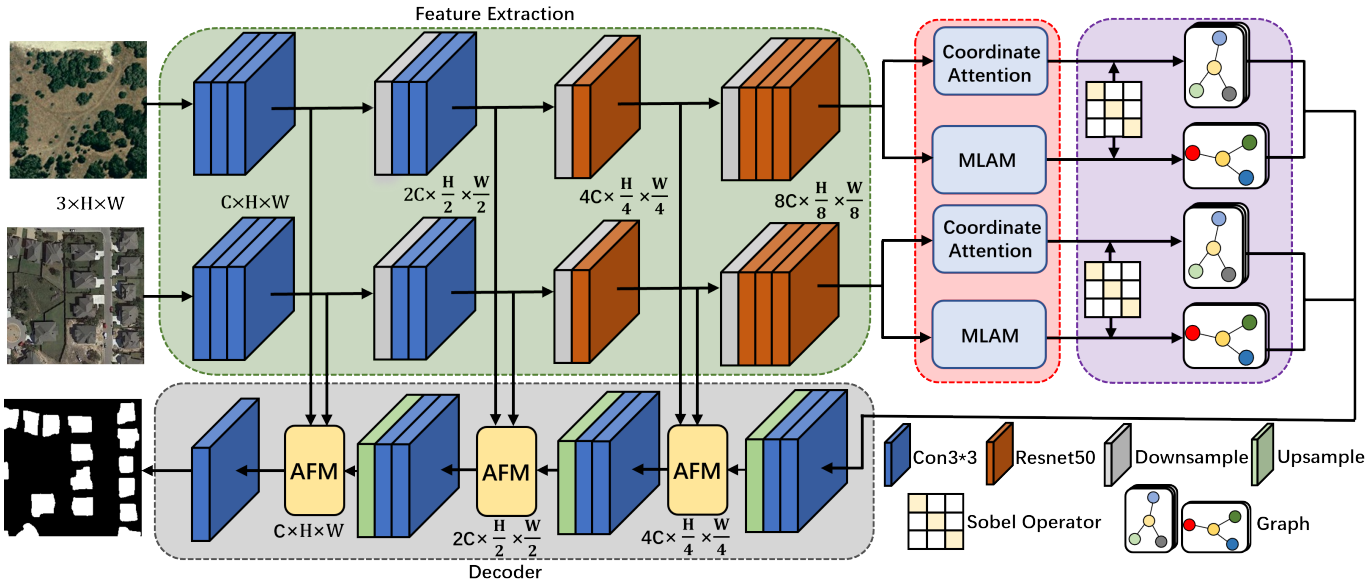


Fig. 1. Proposed ASGCN for CD, which contains four parts: feature extraction module, DAM, Sobel GCN module, and decoder with AFM.

Given the input images X_1, $X_2 \in \mathbb{R}^{3 \times H \times W}$, we first use a convolution operation with a kernel size of 3×3 to ascend dimension from 3 to C, where $C = 64$. Then, after other two convolution layers with a kernel size of 3×3, each layer of the convolution operations is followed by a batch normalization (BN) and a rectified linear unit (RELU) activation function. The size of the feature map will be $\mathbb{R}^{C \times H \times W}$. Then we use a convolution operation with a kernel size of 3×3 and stride of 2 to downsampling the scale of the feature map, as the same, followed by three convolution layers with a kernel size of 3×3; the size will be $\mathbb{R}^{2C \times (H/2) \times (W/2)}$.

To obtain multiscale features, CNN architecture is used in this study. The current classical CNN architectures are AlexNet [35], GoogleNet [36], VGG [37], and ResNet [38]. As the network depth increases, the network becomes less and less effective, and even network degradation occurs. ResNet is a very important framework in semantic segmentation, which uses residual blocks as the basic blocks of its architecture, and the introduction of a shortcut connection can solve the problems of gradient explosion and gradient disappearance. The bottleneck residual block in ResNet-50 [38] is a type of the residual block that uses 1×1 convolution to create a bottleneck, which reduces the parameters and matrix multiplication number and reduces the training time. It uses 1×1 convolution to group the input channels and convolves the grouped input channels with the grouped output channels to reduce the computation and accelerate the model inference without affecting the model performance.

We use bottleneck residual block of ResNet-50 in the encoder for feature extraction, divided into four groups, which contain 3, 4, 6, and 3 bottlenecks, respectively. We keep the bottleneck structure from ResNet-50, but only perform the sampling operation before the first two groups. The remaining two groups do not change the image size, which preserves more detailed information and reduces the computation. Before the first group of ResNet-50, we use max-pooling operation for downsampling. And a 3×3 convolution layer with the stride of 2 is added before the second group to keep downsampling. After the rest three groups of ResNet-50, we get high-dimensional features. In the encoder, we obtain four different scales of feature maps with rich feature information. Afterward, it will be fused with the decoder features. The final feature map is fed into the subsequent network for further processing.

In the feature extraction module, we obtained information at different levels and scales by extracting features at each level to better capture the contextual information and regional features of the object.

B. Dual-Branch Attention Module

After feature extraction, we split the input into two branches, separately entering two attention modules: coordinate attention [39] and MLAM, which focus on spatial and channel information.

In previous studies, there are many researchers applying the attention mechanism to the field of CD, for example, Chen et al. [42] proposed axial attention, which is in the decoder, superimposes the multiscale features of the encoder and decoder, takes the concatenated feature as an input, and performs feature enhancement in the direction of the X- and Y-axes. Our proposed attention model, in addition to feature enhancement of the multiscale information in the coordinate point of view, takes into account the localized information and pays attention to a broader scope. Ding et al. [43] proposed DSA-Net, which uses a deeply supervised attention-guided network for building CD global, and it uses two convolutional branches to extract the deep features and the shallow features, respectively, and uses a spatial attention model to encode the importance of each pixel location. In contrast, our proposed multiscale attention model uses more convolutional kernels of different sizes to extract more detailed features, and we

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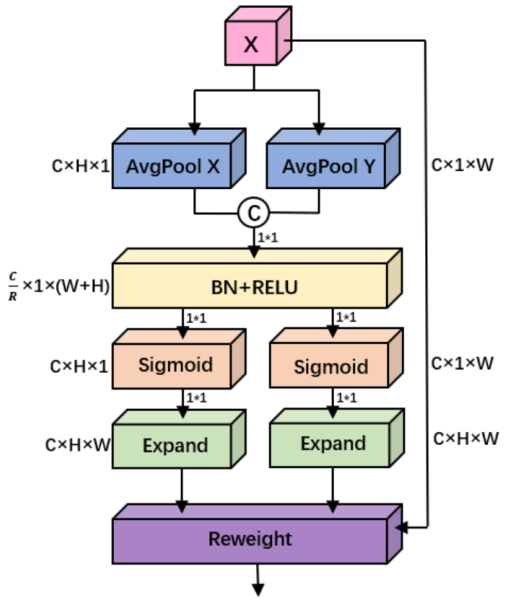


Fig. 2. Coordinate attention.

also pay more attention to local information when encoding the importance of each pixel location. MASNet [44] applies PAM to four different scales of features extracted by the feature extraction module and stitched together; we differ from it by actively using a multiscale convolutional kernel for fine-grained feature extraction and the proposed MLAM for feature refinement.

In the following, we will explain these two attention modules in detail.

1) Coordinate Attention: First, coordinate attention [39] is applied for feature refinement, which takes into account how channel information and location information are related in the feature space as shown in Fig. 2.

Given the input $X \in \mathbb{R}^{C\times H\times W}$, it uses a 1-D horizontal global pooling and a 1-D vertical global pooling to encode each channel along the horizontal and vertical coordinates, respectively, with the size of $C \times 1 \times W$ and $C \times H \times 1$, and then we concatenate them and use 1×1 convolution for transformation, formulated as follows:

$$X' = \delta(\mathrm{Con}_{1\times1}([\mathrm{avgpool}x(X), \mathrm{avgpool}y(X)])) \tag{1}$$

where $[\cdot,\cdot]$ denotes the concatenation operation, δ means a nonlinear activation function, and $\mathrm{Con}_{1\times1}$ represent 1×1 convolution and $X' \in \mathbb{R}^{(C/r)\times1\times(W+H)}$.

It then splits the tensors along the spatial dimension into two separate tensors: $X_h \in \mathbb{R}^{(C/r)\times H}$ and $X_w \in \mathbb{R}^{(C/r)\times W}$. Another two 1×1 convolutional operations separately transform X_h and X_w to the same channel number of the input X, and the outputs are then expanded and used, correspondingly, as attention weights.

2) Multiscale Local Attention Module: On the other branch, we propose an attention module with multiscale expression capability and focus on local features called MLAM as shown in Fig. 3. For the input feature map $X \in \mathbb{R}^{C\times H\times W}$. First, we transform the input features with channel C into four branches with channel $(C/4)$. To achieve separation, we use four convolution with different sizes of kernels to obtain different scales of information. Specifically, the sizes of the four convolution kernels are 3×3, 5×5, 7×7, and 9×9. The formula for the separation is as follows:

$$X_1 = [C_{3\times3}(X), C_{5\times5}(X), C_{7\times7}(X), C_{9\times9}(X)] \tag{2}$$

where $[\cdot,\cdot]$ denotes the concatenation operation, and $C_{i\times i}$ means the convolution operation with the kernel size of i $(i = 3, 5, 7, 9)$. Each convolution operation will change the number of channels to $(C/4)$. Independent learning of multiscale spatial information takes place for each branch, and then we concatenate the four branches and build a cross-channel interaction. The fully connected layer and global average pooling are used to get attention map in the direction of a channel, the four-channel attention map is stitched together, and Soft-max is performed to calibrate. Then we multiply the channel attention weight with the initial feature map to get the first-stage features. Thus, we obtain images rich in multiscale information.

After that, we have worked on the spatial orientation as well; given the channel attention features X_C as input, we first use a local maximum pooling and local average pooling with kernel size and a stride of 2 to aggregate the features and generate two spatial descriptors X_M, $X_A \in \mathbb{R}^{C\times(H/2)\times(W/2)}$, and the calculations are as follows:

$$X_M = \mathrm{MaxPooling}_{k,s=2}(X_C) \tag{3}$$

$$X_A = \mathrm{AvgPooling}_{k,s=2}(X_C) \tag{4}$$

and then we use global maximum pooling and global average pooling with kernel size and a stride of 2 in the direction of channel to further compress the descriptors X'_M, $X'_A \in \mathbb{R}^{1\times(H/2)\times(W/2)}$, calculated as follows:

$$X'_M = \mathrm{GobalMaxPooling}_{k,s=2}(X_M) \tag{5}$$

$$X'_A = \mathrm{GobalAvgPooling}_{k,s=2}(X_A) \tag{6}$$

and then concatenate them into a single spatial descriptor operator as subsequent input to save computation. Afterward, we apply a dilated convolution with a dilation of 2 to increase the receptive field and reduce the feature loss brought on by multiple-step pooling operations and enhance the nonlinear characterization capability of spatial descriptors. In the end, we use nearest neighbor interpolation to upsample and get the spatial attention map, and the calculations are as follows:

$$X_{L1} = \sigma\left(\mathrm{Up}\left(\mathrm{Con}_{k=3,d=2}\left(\left[X'_M, X'_A\right]\right)\right)\right) \tag{7}$$

where $[\cdot,\cdot]$ denotes the concatenation operation, $\mathrm{Con}_{k=3,d=2}$ means the convolution operation with the kernel size of 3 and dilation of 2, Up means the nearest neighbor interpolation, and σ means the Sigmoid activation function.

With the DAM, the network is able to better focus on the informative features of the target object, thus improving the detection of the target object. This can help reduce false detections and improve the accuracy of small targets' detection.

C. GCN Module

In GCN, an advanced deep learning network, the aggregation of image neighborhood features is realized by graph

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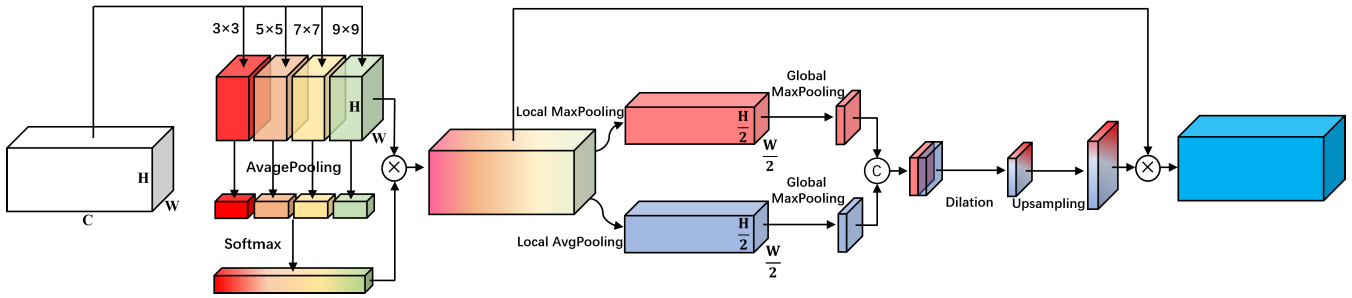


Fig. 3. MLAM, where attention weights are calculated separately in both spatial and channel directions.

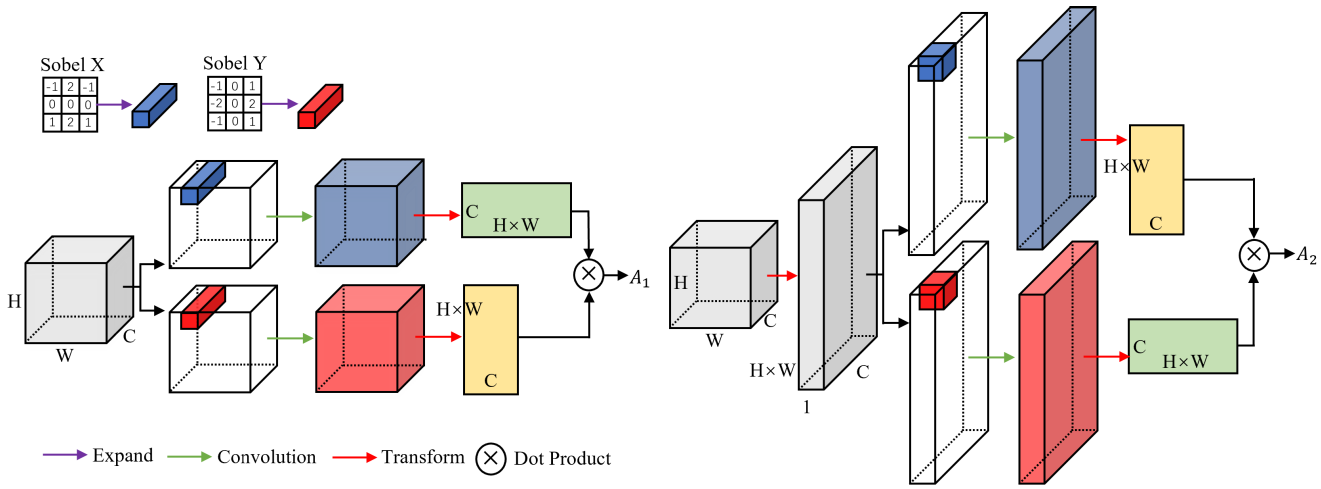


Fig. 4. Construction of adjacency matrix, using Sobel operator to construct adjacency matrices in 2-D separately. We obtain adjacency matrices A_1 and A_2 at different scales.

nodes, which can fully extract the features in VHR and effectively maintain the category boundary information. CNN cannot handle non-Euclidean structure data; it cannot maintain translation without deformation on non-Euclidean structure data; in GCN, vertices and edges are used to build topology graphs with the corresponding relationships and extract spatial features of irregular topology graphs. Such a feature makes the use of GCN have certain advantages.

In this article, we construct two scales of adjacency matrices by a gradient operator with different scales, focusing on spatial and channel information, respectively.

We define the relationship of spectral features in an VHR as an undirected graph of $G = (V, E)$, where V means the node set with $V = V_1, V_2, \ldots, V_N$, and E denotes the edge set. We can define the adjacency matrix as $A \in R^{N \times N}$, describing the internal relationship between nodes. The propagation equation of the graph convolution is as follows:

$$H^{l+1} = \sigma\left(\widetilde{D}^{-\frac{1}{2}}\widetilde{A}\widetilde{D}^{-\frac{1}{2}}H^l W^l\right). \tag{8}$$

In GCN, it is important how the adjacency matrix is constructed; in Graph-FCN [40], each pixel position in the feature layer is treated as a node, and the node is connected with its neighboring nodes and also use the Gauss kernel function. AGCN [41] used the generalized Mahalanobis distance as a metric; in MSSGU [47], if two superpixels are adjacent, they consider them to be adjacent and create an edge between them.

Wu et al. [45] proposed a model in which the image is first subjected to advanced feature extraction and multiscale segmentation, and then the average over all the scales of each segmented object is used as the object node to construct the graph. In SIGNet, proposed by Zhou et al. [46], the graph convolution operation is performed on the visual features of images and the semantic representation of categories and the adjacency matrix of the nodes is updated through gradient descent during training. In our network, we compute the edge information using the edge gradient operator; we use the Sobel operator to construct a graph convolution adjacency matrix and encode this edge information into the adjacency matrix.

The proposed module constructs an adjacency matrix using a gradient operator. The Sobel operator is an edge detection algorithm, which is widely used in image processing and computer vision to extract edge information from images, and the resulting image is an edge intensity image. We use the Sobel operator to construct a graph convolution adjacency matrix to convert the original image into an edge intensity image and encode this edge information into the adjacency matrix, which enables the GCN to more effectively capture the image's local structural information. Compared with traditional image convolution networks, this approach is more advantageous for remote sensing image CD, particularly when we concentrate on the image's edge characteristics.

It is widely used in digital image processing because of its simplicity of implementation and computational efficiency.

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It contains two sets of 3×3 matrices, horizontal and vertical, which are performed convolution operation with the image to obtain the horizontal and vertical brightness difference values, respectively, the location where node values change dramatically contains more information, and it is more reasonable to build edges between these nodes.

As summarized in Fig. 4, we divide the graph construction into two branches. In the first branch, we expand the channels of the gradient operator Sobel X and Sobel Y matrices into C, get convolution operation in the input map as convolution kernels, transform the obtained features in size $\mathbb{R}^{C \times H \times W}$ to obtain two matrices of size $C \times N$ and $N \times C$, where $N = H \times W$, and dot product the matrices to obtain the adjacency matrix $A_1 \in C \times C$, In the other branch, we first transform the input map by dimension to $\mathbb{R}^{1 \times D \times C}$, where $N = H \times W$, and get convolution operation in the 1-D with matrices Sobel X and Sobel Y as convolution kernels on the features. Similarly, the obtained features are dimensionally transformed to obtain the matrices $N \times C$, $C \times N$, and the matrices are dot producted to obtain the adjacency matrix $A_2 \in N \times N$.

After constructing the adjacency matrix, propagation is performed as shown in (8), after the construction and learning of the graph, feature map is more detailed. CD problem requires considering the differences between the target and its surroundings, and these differences are usually local. This module is able to capture contextual information, combining information from the surrounding environment with target features for better CD.

D. Decoder and AFM

After feature extraction module, DAM, and GCN module, we get two branches of refined features, the size of each branch is $\mathbb{R}^{8C \times H \times W}$, and we concatenate them and send them into the decoder. First, we use two convolution layers with the kernel size of 3 to reduce the dimension, followed by an upsampling layer with a kernel size of 3 and a stride of 2 to increase the size and decrease the dimension at the same time; we repeat these three layers three times in the decoder, each time outputting into AFM to fuse with the feature maps of the encoder at different scales.

Skip connections are often used in residual networks to solve the problem of gradient explosion and gradient disappearance during training process in deeper networks. Feature fusion means the combination of features from different layers or branches, which is a ubiquitous component of network. It is usually implemented by summation or concatenation, but this may not be the best option. In computer vision, attention mechanisms allow features to focus more on informative areas. It is a technique that help a model focus on the most important regions of an image. Specifically, the attention mechanism can focus the model's attention on the regions of higher importance by weighting different parts of the image. This design improves the accuracy and efficiency of the network and makes the model more adaptable to complex and changing real-world scenarios.

Convolutional block attention module (CBAM) [48] is a classic attention module, which performs attention operations in terms of both the spatial and channel dimensions; we fuse the bitemporal image in the encoder with the output in the decoder to obtain the changing pixels more efficiently and propose a fusion module with attention mechanism that more effectively fuses features while focusing on prechange and postchange features.

As shown in Fig. 5, the output features of the two branches of the encoder X_1 and $X_2 \in \mathbb{R}^{2^i C \times (H/2^i) \times (W/2^i)}$ (i denotes i_{th} stage) are concatenated, whose size is $\mathbb{R}^{2^{i+1} C \times (H/2^i) \times (W/2^i)}$, and then go through two convolution layers with a kernel size of 3×3 to reduce dimension from 2^{i+1} to 2^i; each layer is followed by a BN and a RELU activation function. The features of decoder $Y \in \mathbb{R}^{2^i C \times (H/2^i) \times (W/2^i)}$ have a fine-grained character, so it is inputting to the fusion module to integrate with the features of the encoder. After concatenation, we use the spatial and channel attention CBAM [48] to make fusion features spatially more focused on effective information and enable more efficient integration of information in the direction of the channel. Finally, two convolution layers with a kernel size of 3×3 are applied to reduce dimension from 2^{i+1} to 2^i, and the size of output feature map is $\mathbb{R}^{2^i C \times (H/2^i) \times (W/2^i)}$.

IV. Experiments

A. Datasets

On three difficult datasets, we conducted comparative experiments to assess the performance and robustness of ASGCN, including LEVIR-CD datasets, WHU-CD datasets, and DSIFN-CD datasets.

LEVIR-CD [25] comprises 637 image pairs, each of which is 1024×1024 pixels. Most of the images in this dataset focus on changes in buildings, annotated 31 333 changed buildings in the label data. The dataset was partitioned into training, testing, and validation sets at random by the authors. We use the cropped data with the size of 256×256 pixels without overlapping; the dataset consists of 2406 pairs for training, 731 pairs for testing, and 333 pairs for validation, respectively.

The WHU-CD [54] dataset contains two remote sensing images with 0.075-m spatial resolution; the images were taken in Christchurch, which had a magnitude of 6.3 earthquake in 2011. This dataset captured the city's postearthquake reconstruction situation. In this dataset, the changes need to detect only include the changes caused by buildings; we crop them into 224×224 nonoverlapping image patches. And in an 8:1:1 ratio, randomly divided these images into a training set, validation set, and testing set.

The DSFIN-CD [55] dataset was collected from Google Earth through manual effort. The dataset contains six large bitemporal high-resolution images, which are six cities in China. Some of them have been cropped into 394 pairs of 512×512 subimages. And after data augmentation, 3940 bitemporal image pairs are obtained. For the testing set, 48 image pairs are created from the Xian image pair. The training dataset has 3600 image pairs, whereas the validation dataset contains 340 image pairs. We use nonoverlapping patches of size 256×256 of the data.

B. Implementation Detail

The environment and framework we use are Python v3.7 and PyTorch 1.1.0, respectively. We implement our model using

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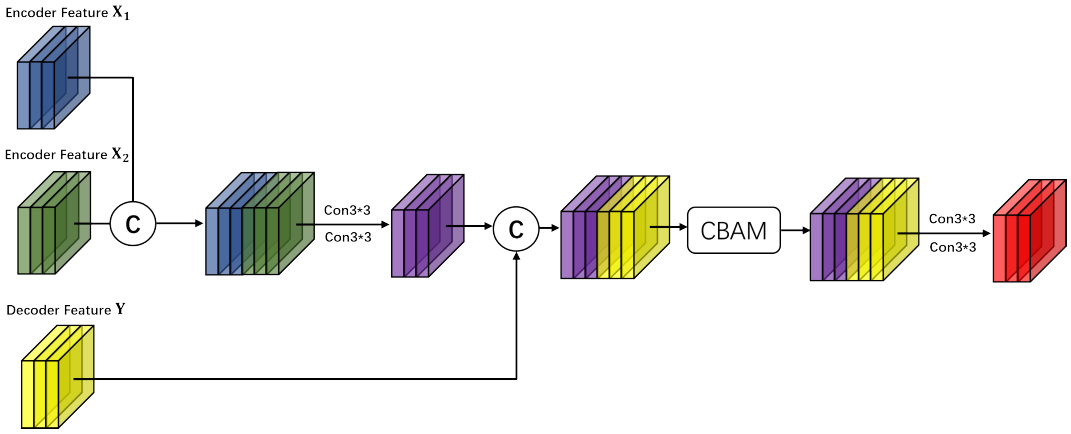


Fig. 5. AFM, which fuses feature maps in the direction of the channel without changing the resolution.

TABLE I
COMPARISON RESULTS ON THE THREE CD TEST SETS. THE HIGHEST SCORE IS MARKED IN BOLD. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%)

	LEVIR-CD F1 / IoU / Pre. / Rec. / OA	**WHU-CD** F1 / IoU / Pre. / Rec. / OA	**DSIFN-CD** F1 / IoU / Pre. / Rec. / OA
FC-EF [50]	83.40 / 71.53 / 86.91 / 80.17 / 98.39	69.37 / 53.11 / 71.63 / 67.25 / 97.61	61.09 / 43.98 / 72.61 / 52.73 / 88.59
FC-Siam-Di [50]	86.31 / 75.92 / 89.53 / 83.31 / 98.67	58.81 / 41.66 / 47.33 / 77.66 / 95.63	62.54 / 45.50 / 59.67 / 65.71 / 86.63
FC-Siam-Conc [50]	83.69 / 71.96 / 91.99 / 76.77 / 98.49	66.63 / 49.95 / 60.88 / 73.58 / 97.04	59.71 / 42.56 / 66.45 / 54.21 / 87.57
DTCDSCN [29]	87.67 / 78.05 / 88.53 / 86.83 / 98.77	71.95 / 56.19 / 63.92 / 82.30 / 97.42	63.72 / 46.76 / 53.87 / 77.99 / 84.91
STANet [25]	87.26 / 77.40 / 83.81 / **91.00** / 98.66	82.32 / 69.95 / 79.37 / 85.50 / 98.52	64.56 / 47.66 / 67.71 / 61.68 / 88.49
IFNet [55]	88.13 / 78.77 / **94.02** / 82.93 / 98.87	83.40 / 71.52 / **96.91** / 73.19 / 98.83	60.10 / 42.96 / 67.86 / 53.94 / 87.83
SNUNet [51]	88.16 / 78.83 / 89.18 / 87.17 / 98.82	83.50 / 71.67 / 85.60 / 81.49 / 98.71	66.18 / 49.45 / 60.60 / 72.89 / 87.34
BIT [52]	89.31 / 80.68 / 89.24 / 89.37 / 98.92	83.98 / 72.39 / 86.64 / 81.48 / 98.75	69.26 / 52.97 / 68.36 / 70.18 / 89.41
ChangeFormer [53]	90.40 / 82.48 / 92.05 / 88.80 / 99.04	88.42 / 79.24 / 93.55 / 83.82 / 99.13	86.67 / 76.48 / **88.48** / 84.94 / 95.56
ASGCN(ours)	**91.60** / **84.51** / 92.94 / 90.31 / **99.16**	**91.26** / **83.93** / 92.42 / **90.14** / **99.32**	**88.02** / **78.61** / 87.73 / **88.33** / **95.92**

PyTorch and use an NVIDIA RTX 3090 GPU for training. In the process of training, we apply data augmentation through random flip, random rescale (0.8–1.2), random crop, and Gaussian blur. We train the models with a weight decay of 0.01 and beta values of (0.9, 0.999) using the cross-entropy (CE) loss and AdamW optimizer. The learning rate starts off at 0.0001 and declines linearly to 0. The training periods are 200 epochs. To train the model, we chose a batch size of 8.

C. Performance Metrics

We use intersection over union (IoU) and $F1$ scores of the relevant change categories as the main evaluation metrics. The precision and recall of the test are used to generate $F1$ score as follows:

$$F1 = \frac{2}{\text{recall}^{-1} + \text{precision}^{-1}} \tag{9}$$

$$\text{IoU} = \text{TP}/(\text{TP} + \text{FN} + \text{FP}). \tag{10}$$

In addition, precision, recall, and overall accuracy (OA) are also calculated. These metrics are described as follows:

$$\text{recall} = \text{TP}/(\text{TP} + \text{FN}) \tag{11}$$

$$\text{precision} = \text{TP}/(\text{TP} + \text{FP}) \tag{12}$$

$$\text{OA} = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FN} + \text{FP}) \tag{13}$$

where TP, FP, and FN represent the number of true positive, false positive, and false negative, respectively.

D. Comparison to State-of-the-Art

In this part, we contrast the CD performance of our ASGCN-CD with that of other SOTA approaches already in use.

1) FC-EF [50] is a method that uses bitemporal images, which are concatenated as a single input and go through a ConvNet for detecting.
2) FC-Siam-Di [50] is a method with multiscale feature and uses features extracted by bitemporal FCN and absolute difference operation to detect change regions.
3) FC-Siam-Conc [50] is a multiscale feature method, which extracts multilevel features by a Siamese FCN and uses the concatenation of features to detect changes.
4) DTCDSCN [29] is a CD method with attention mechanism and uses attention module to extract the interdependencies of features to detect changes.
5) STANet [25] is a CD method with Siamese-based spatial–temporal attention network, which uses attention mechanism to capture multiscale features.
6) IFNet [29] is a method with multiscale feature and fuses multilevel deep features and image difference features for change map reconstruction by attention module.
7) SNUNet [51] is a method for concatenating multiple levels of features that uses NestedUNet to detect changes.
8) BIT [52] is a CD method with transformer, which uses a encoder–decoder network with transformer to use

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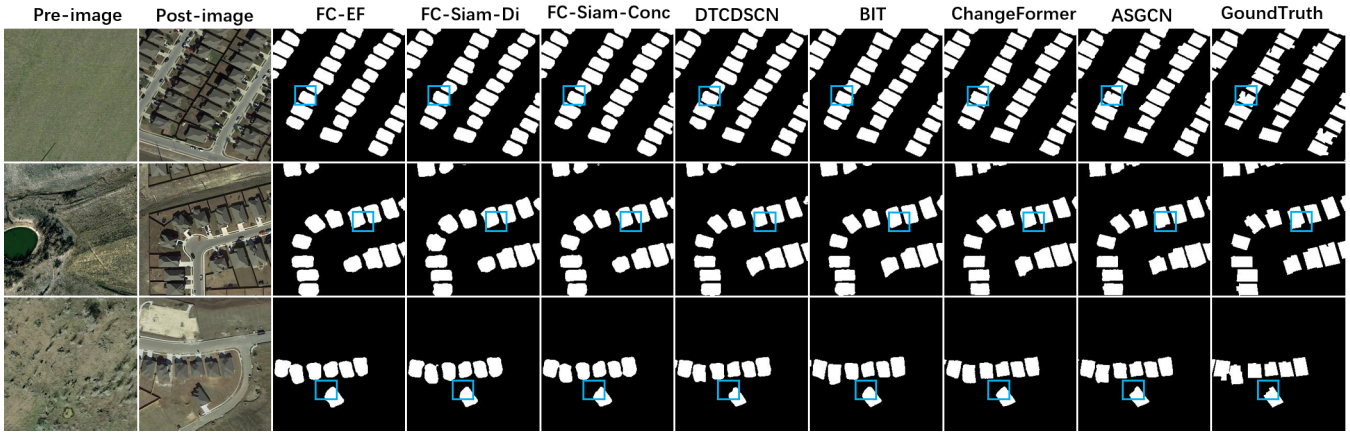


Fig. 6. Visualization comparison. As highlighted in blue, our ASGCN captures more edge details, compared with other SOTA methods.

TABLE II

ABLATION STUDY OF THE PROPOSED THREE MODULES: DAM, GCN MODULE, AND AFM. WE PERFORM DIFFERENT COMBINATIONS WITH $F1$, IOU, AND OA SCORES ON LEVIR-CD DATASET

DAM	GCN	AFM	F1	IoU	OA
			90.22	82.18	99.01
✓			90.71	82.99	99.06
	✓		91.03	83.54	99.09
		✓	90.70	82.92	99.05
✓	✓		91.26	83.93	99.12
✓		✓	91.03	83.53	99.10
	✓	✓	91.28	83.95	99.13
✓	✓	✓	**91.60**	**84.51**	**99.16**

semantic tokens to enhance the context information of ConvNet features and uses feature differencing to detect changes.

9) ChangeFormer [53], a Siamese network architecture with transformer, combines hierarchically transformer encoder with multilayer perception (MLP) decoder for feature extracting to obtain change map.

We shows the results of the existing CD methods on the test sets of the LEVIR-CD, WHU-CD, and DSIFN-CD datasets. As shown in Table I, the proposed ASGCN-CD network improves performance across a variety of $F1$, IoU, recall, and OA. More specifically, our ASGCN-CD improves in $F1$/IoU/OA by 1.2/2.03/0.12%, 2.86/4.69/0.19%, and 1.35/2.13/0.36% for LEVIR-CD, WHU-CD, and DSIFN-CD datasets, respectively. On the experiments with the three datasets, our method achieves the highest precision in all the three metrics of $F1$ IOU OA and also obtains the highest recall precision on the DSIFN-CD and WHU-CD datasets, and due to the way the metrics are computed, the precision is relatively lower when recall is high, and in the field of CD research, the computed metrics of $F1$ and IOU are more meaningful.

In addition, Fig. 6 analyzes the visual excellence of several CD techniques using the LEVIR-CD dataset. The details of change map are highlighted in blue. Compared with other SOTA methods, our ASGCN-CD can capture more accurate details. These quantitative and qualitative experiments and comparisons show that our proposed CD method has advantages over the existing SOTA method.

TABLE III

ABLATION STUDY OF THE PROPOSED ATTENTION MODULES. WE CONSTRUCT THE CONVOLUTION OF THE GRAPH IN DIFFERENT DIRECTIONS AND GET $F1$, IOU, AND OA SCORES ON LEVIR-CD DATASET

CA	MLAM	F1	IoU	OA
		91.28	83.95	99.13
✓		91.32	84.06	99.14
	✓	91.41	84.19	99.14
✓	✓	**91.60**	**84.51**	**99.16**

TABLE IV

ABLATION STUDY OF THE PROPOSED GCN MODULE. WE CONSTRUCT THE CONVOLUTION OF THE GRAPH IN DIFFERENT DIRECTIONS AND GET $F1$, IOU, AND OA SCORES ON LEVIR-CD DATASET

A1	A2	F1	IoU	OA
		91.03	83.53	99.10
✓		91.14	83.73	99.11
	✓	91.36	84.09	99.14
✓	✓	**91.60**	**84.51**	**99.16**

E. Ablation Studies

We perform ablation experiments on the three modules designed to test their performance in our CD network. Ablation experiments were performed on the LEVIR-CD dataset, by conducting the experiments while ignoring various modules and compare the importance of each module for the CD task. The results are shown in Table II; the highest increase in accuracy is obtained when applying the GCN module alone, with a 1.33% increase in IoU and 0.81% increase in $F1$ score. When two of the modules are applied, the best accuracy is obtained by combining the DAM and GCN modules, with an increase of 1.75% in IoU and 1.04% in $F1$ score. From the table, it can be seen that the best experimental results and best accuracy are obtained when these three modules are combined together.

In DAM, we experimented with an attention module alone to determine the effect of two attention modules on accuracy. As can be seen from Table III, each individual attention module can improve accuracy to some extent. But spatial and channel features are better emphasized and work best when both the attention modules are used simultaneously.

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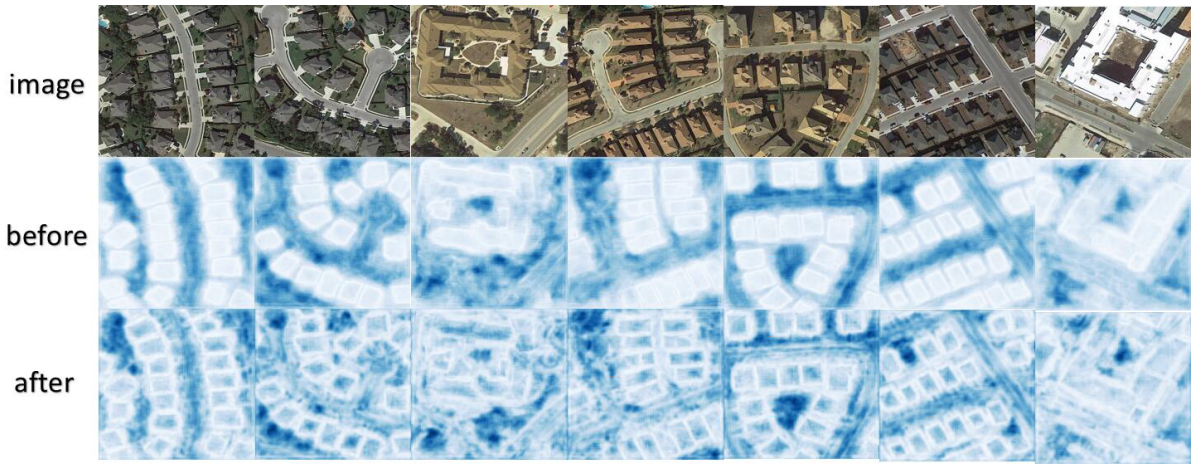


Fig. 7. Feature visualization, including extracted feature maps with and without SobelGCN.

In the GCN module, we construct two adjacency matrices A_1 and A_2 in different directions using the Sobel operator. To demonstrate the effect of the two adjacency matrices, we conducted experiments using one of the adjacency matrices separately, and the experimental results in Table IV show that the accuracy is improved when the adjacency matrix in one direction alone is used, but the effect is not obvious, and the highest accuracy of CD is achieved when both the adjacency matrices are used simultaneously in both the directions, maximizing the focus on edge information.

As can be seen in Fig. 7, we extracted the feature maps with and without the proposed SobelGCN, and the results show that the feature maps after SobelGCN pay more attention to the edge information, including the edges of the houses, and this information plays a role in the subsequent CD processing.

V. Conclusion

In this article, we propose ASGCN for CD. The structure is divided into feature extraction module, DAM, GCN module, AFM, and decoder; it fully uses the strengths of CNN and GCN, which is capable of extracting adequate features from multitemporal VHR and focuses on expanding differences in features to facilitate CD.

In GCN module, we use Sobel operator to conduct graph, which can enlarge the receptive field of feature and effectively preserves the class boundary, which improve the accuracy of CD. In DAM and AFM, we use attention mechanism for feature refinement in both the spatial and channel directions, focusing more on informative pixels and get better results obtained in networks combined with GCN module. On three CD datasets, compared with other ways, we perform more effectively in terms of accuracy and efficiency.

Overall, our proposed method solves problems such as lack of edge information in CD to some extent and improves the accuracy of detection. However, the current method acquires edge information and global context information in a single way. With the increasing computational power and the continuous development of deep learning algorithms, it can be expected that more efficient and accurate deep learning models will be developed for remote sensing image CD. Future research may explore how to effectively fuse remote sensing image data from multiple sensors with multiple resolutions to improve the accuracy and robustness of CD. Meanwhile, future research may explore more semi-supervised learning and unsupervised learning methods to reduce the need for labeled data and improve the generalization ability of the models.

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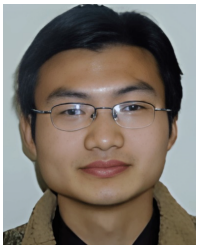
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